Stuti Nayak

The Hague, Netherlands | stuti257@gmail.com | stutinayak.com | GitHub | LinkedIn

Applied ML professional with 4+ years of industry experience and additional academic research in machine learning, data pipelines and model evaluation. Built transformer-based NLP and speech classifiers for noisy real-world data, including end-to-end experimentation and cloud inference.

Work Experience

Data Engineer, Aurelius Enterprise, Amsterdam, Netherlands April 2025 - Present

- Support a confidential client-facing AI automation initiative by examining source structures, preparing data and recreating operational KPIs to test technical feasibility.
- Design evaluation criteria for AI outputs, defining success measures, data gaps, failure modes and stakeholder usability before model development.
- Designed and built production event and telemetry ingestion pipelines, transforming noisy upstream data into schema-validated datasets for analytics and downstream processing.
- Worked with client and engineering stakeholders on data quality rules, dataset contracts, lineage and governance, improving trust in downstream data consumption.
- Contributed feature development and bug fixes to an open-source metadata platform; documented integration decisions and technical handovers for maintainable data services.

Data Scientist, Knownwell, The Hague, Netherlands February 2024 - July 2024

- Developed a BERT/BERTopic-based classification and topic-routing model, improving match accuracy from 20% to 30% on real-world text data.
- Built and validated a separate RoBERTa sentiment classifier, applying held-out evaluation and error analysis to understand model quality and limitations.
- Translated an ambiguous stakeholder question into a measurable ML problem with defined metrics, repeatable comparisons and actionable model evaluation.
- Iterated on NLP preprocessing, inference prototypes and evaluation workflows while communicating technical trade-offs to non-technical stakeholders.

Data Scientist, Rate.nl, Rotterdam, Netherlands August 2021 - September 2023

- Built an audio classification pipeline using wav2vec 2.0 on noisy real-world call recordings, improving accuracy from 27% to 55% through preprocessing and experimentation.
- Designed and ran model experiments, tracked metrics with MLflow and analyzed failure cases to choose a reliable production candidate.
- Packaged and deployed containerized model inference with Docker on Google Cloud Run, translating model research into a usable cloud service.
- Built a Gradio application with model-output visualizations so stakeholders could inspect predictions and provide feedback during rollout.

Researcher, Clinical Data Science, Maastricht University, Netherlands September 2019 - August 2020

- Configured federated learning infrastructure for collaborative cross-hospital analytics, supporting privacy-preserving experimentation across distributed data sources.
- Designed a semantic model to standardize data across hospitals and reduce preprocessing overhead in collaborative research.

Research Assistant, Institute of Data Science, Maastricht University, Netherlands July 2017 - February 2019

- Built a deep learning NLP pipeline over scientific abstracts for biomedical metadata quality assessment, cleaning and preparing text for modelling.
- Developed CNN and BiLSTM multi-label classifiers, with the CNN achieving 83% accuracy on held-out evaluation.

Core Skills

Deep Learning & Representation Learning: Python, PyTorch, TensorFlow, Hugging Face Transformers, BERT, RoBERTa, wav2vec 2.0, CNN, BiLSTM, model training, feature representations

ML Evaluation & Experimentation: held-out validation, cross-validation, precision/recall, classification accuracy, imbalanced learning, error analysis, statistical comparison, model selection, MLflow

Production ML & Software: scikit-learn, Docker, Google Cloud Run, Git, GitHub Actions, CI/CD, reproducible experiments, cloud inference, Gradio, maintainable Python

Data Processing & Pipelines: pandas, NumPy, SciPy, SQL, PySpark, Kafka, event and telemetry data, ETL/ELT, feature engineering, schema validation, data quality, BigQuery, PostgreSQL

Cloud & Applied AI: Google Cloud Platform, Azure, RAG, LangChain, FAISS, semantic search, confidence-based routing, experiment reporting, stakeholder-facing visualizations

Selected Projects

- **Credit Card Fraud Detection (personal project):** Evaluated a LightGBM classifier on 284,807 highly imbalanced transactions; used temporal splitting, threshold optimization, ROC-AUC around 0.98 and SHAP to analyse model behaviour.
- **RAG & Semantic Search Q&A (personal project):** Built a retrieval pipeline using LangChain, FAISS, document chunking, semantic similarity and confidence-based fallback, focusing on output reliability.
- **Customer Churn & Retention Analytics (personal project):** Used SQL and behavioural feature engineering to identify at-risk customer segments and make prediction outputs actionable.

Education and Certifications

Maastricht University | M.Sc. in Data Science in Decision Making | Maastricht, Netherlands

Dhirubhai Ambani University | B.Tech. in Information and Communication Technology | Gandhinagar, India

Certifications: Confluent Certified Developer for Apache Kafka; Certified Data Management Professional - Associate; Databricks Certified Data Engineer Associate in progress